mediated gene transfer. Direct gene transfer includes chemical mediated gene transfer, micro injection, electroporation. Gene gun method and Liposome mediated method of gene transfer. Indirect or vector mediated gene transfer is a method of gene transfer with the help of a plasmid vector. In this method Ti-plasmid from *Agrobactirum tumefeciens* has been used extensively for vector mediated gene transfer.

After the introduction of rDNA into a host cell, it is essential to identify those cells which have received the rDNA molecule. This process is called screening. One of the method of recombinant screening is blue white selection method Replica plating technique in which the pattern of colonies growing on a culture plate is copied. Electrophoresis is a separating technique used to separate different biomolecules.

Blotting techniques are widely used tools for identification of desired DNA or RNA fragments from larger number of molecules. Some of the genetically modified crops are herbicide tolerant – Basta, Dhara mustard, insects resistance – Bt crops, flavrSavr – Tomato, Golden rice. Biopolymers are polyhydroxybutyrate (PHB), polylactic acid (PLA) and green fluorescent protein (GFP) is used to make biosensors. Other applications are biopharming, bioprospecting, biomedication and biofuel, etc.

Evaluation

EHKT6W

1. Restriction enzymes are
   a. Not always required in genetic engineering
   b. Essential tools in genetic engineering
   c. Nucleases that cleave DNA at specific sites
   d. both b and c

2. Plasmids are
   a. circular protein molecules
   b. required by bacteria
   c. tiny bacteria
   d. confer resistance to antibiotics

3. EcoRI cleaves DNA at
   a. AGGGTT b. GTATATC
   c. GAATTC d. TATAGC

4. Genetic engineering is
   a. making artificial genes.
   b. hybridization of DNA of one organism to that of the others.
   c. production of alcohol by using micro organisms.
   d. making artificial limbs, diagnostic instruments such as ECG, EEG etc.,

5. Consider the following statements:
   I. Recombinant DNA technology is popularly known as genetic engineering is a stream of biotechnology which deals with the manipulation of genetic materials by man invitro
   II. pBR322 is the first artificial cloning vector developed in 1977 by Boliver and Rodriguez from E.coli plasmid
   III. Restriction enzymes belongs to a class of enzymes called nucleases.

   Choose the correct option regarding above statements
   a. I & II b. I & III
   c. II & III d. I,II & III

6. The process of recombinant DNA technology has the following steps
   I. amplication of the gene
   II. Insertion of recombinant DNA into the host cells
   III. Cutting of DNA at specific location using restriction enzyme .
   IV. Isolation of genetic material (DNA)

   Pick out the correct sequence of step for recombinant DNA technology.
   a. II, III, IV, I b. IV, II, III, I
   c. I, II, III, IV d. IV, III, I, II

7. Which one of the following palindromic base sequence in DNA can be easily cut at about the middle by some particular restriction enzymes?
   a. 5` CGTTCG 3` 3` ATCGTA 5`

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